ADRIAN SHKUMBOV

adrian.shkumbov@gmail.com || +1 647-939-7310 || www.linkedin.com/in/adrian-shkumbov

Citizenships: Canadian, Bulgarian (*EU*) **Languages:** English (*Native*), French (*Basic*), Bulgarian (*Proficient*)

EDUCATION

University of Toronto 09/2021 – 05/2026

B.A.Sc. in *Engineering Science*, Specialization in *Robotics*

SKILLS

Software: Python (*NumPy, OpenCV, PyTorch*), C, ROS 2, MATLAB, Simulink, Linux, Git

Electrical/Embedded: Nvidia Jetson, ESP32, KiCAD, Oscilloscope, Function Generator, Soldering

Mechanical: SOLIDWORKS, Fusion360, AutoCAD, 3D Printing, Laser Cutting, Basic Machining

EXPERIENCE

Robotics Researcher – Diller Microrobotics Lab, *Toronto, Canada* 09/2025 – Present

- Undertook an undergraduate thesis to improve the operating stability of continuum robots for minimally invasive surgery, developing active compensation control algorithms to preserve procedural safety margins during sudden disturbances.
- Designed and brought-up a compact 10 mm diameter PCB to integrate a high-performance IMU into the distal end of a surgical scale continuum robot.
- Developed a multi-process Python control pipeline integrating ESP32-mediated SPI sensor polling, computer vision tracking, and motor actuation.
- Successfully damped vibrations during robot unjoining through active control, reducing transient tip overshoot from a 25 mm baseline to under 3 mm.
- Co-authoring a technical paper for submission to the IEEE International Conference on Robotics & Automation (ICRA).

Mechatronics Design Engineer – Martinrea International, *Vaughan, Canada* 05/2024 – 09/2025

- Led an emergency on-site deployment overseas to resolve quality complaints from a major auto OEM. Deployed a vision system (Python/OpenCV) on a critical manufacturing cell, devising and integrating a bespoke ML-based inspection system with existing hardware and PLC logic. Resulting system eliminated downtime, with 0 false positives during production.
- Started and led an R&D project to develop a robotic vision system for surface inspection of aluminum castings and automated data collection. Designed (in SOLIDWORKS) a novel multi-camera eye-in-hand module for a collaborative robot (cobot), and manufactured through laser cutting and 3D printing.
- Performed early integration work on cobot module, writing embedded C and ROS2 python nodes for basic functionality. Early inspection trials showed performance comparable to traditional dye penetrant methods, in a fraction of the time.
- Optimized prototypes of an electro-mechanical camera protection device for harsh manufacturing environments. Designed a solenoid driver PCB using KiCAD and redesigned actuating mechanism in SOLIDWORKS. Changes resulted in a 50% reduction in enclosure volume, eliminated thermal concerns, and performed a successful 100,000 actuation cycle stress test.
- Regularly presented work of the Advanced Manufacturing Technology Group during innovation tours for investors, government officials, and company executives.

Engineering Research Intern – University of Strathclyde, *Glasgow, Scotland* 05/2023 – 08/2023

- Developed data fusion programs in MATLAB, integrating GPS coordinates with sonar depth telemetry to generate 3D visualizations of riverbed topology to monitor bridge scour degradation.
- Conducted in-field sonar mapping trials, capturing empirical data to validate the accuracy and viability of low-cost structural health monitoring systems. Submitted technical reports on findings.

Propulsion Sub-team Member – UofT Hyperloop Team, *Toronto, Canada* 09/2022 – 03/2023

- Directed the electrical architecture for a small-scale Linear Halbach Drive, programming embedded C control logic for motor actuation and bench-test validation.
- Designed and 3D-printed specialized component housings in SOLIDWORKS to isolate optical sensors from vibrations and interference during bench tests.